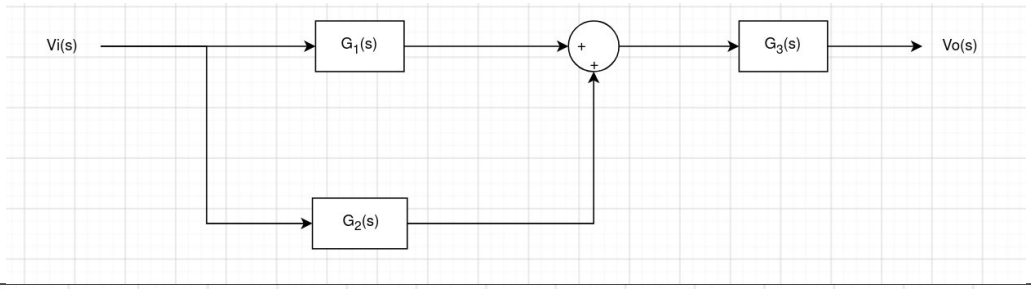
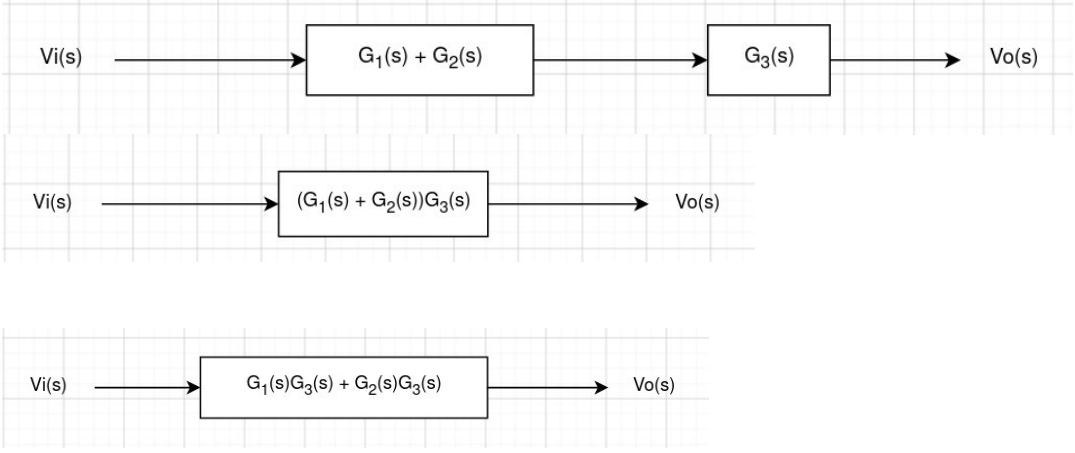
Tugas Pendahuluan Sistem Kendali Najwan I. M NIM: 241354057 Kelas: 2B-EK

1. Apa fungsi diagram blok pada suatu sistem
2. Sederhanakan diagram-diagram blok blok berikut

a.

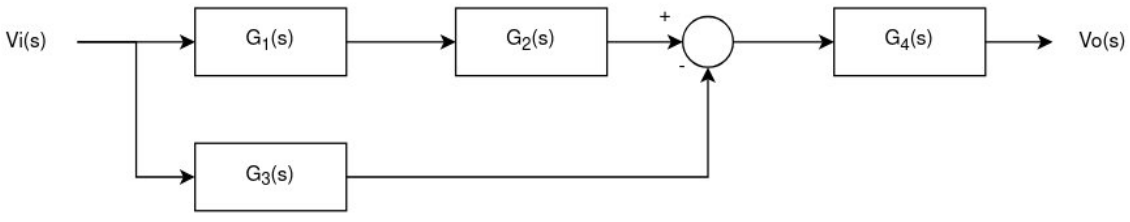


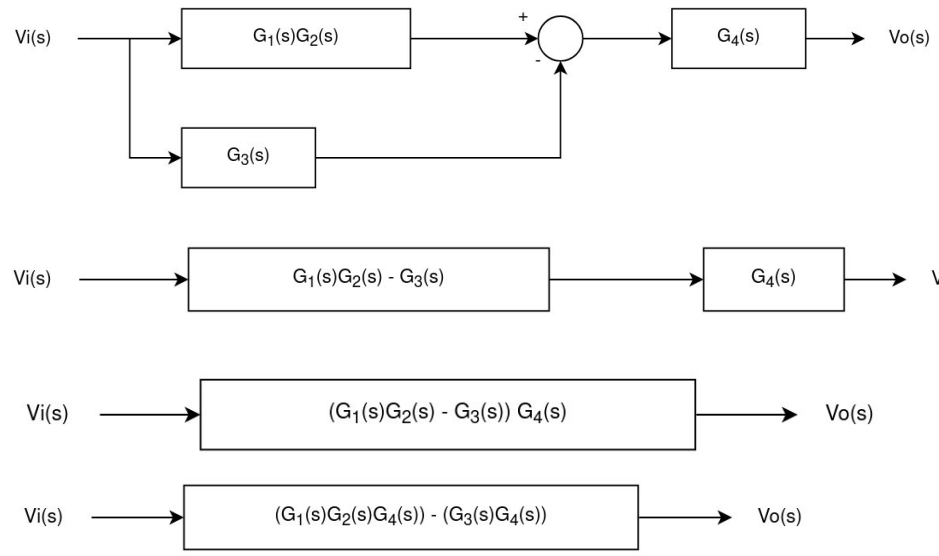


Sehingga:

$$\frac{Vo(s)}{Vi(s)} = G_1(s) \cdot G_3(s) + G_2(s) \cdot G_3(s)$$

b.

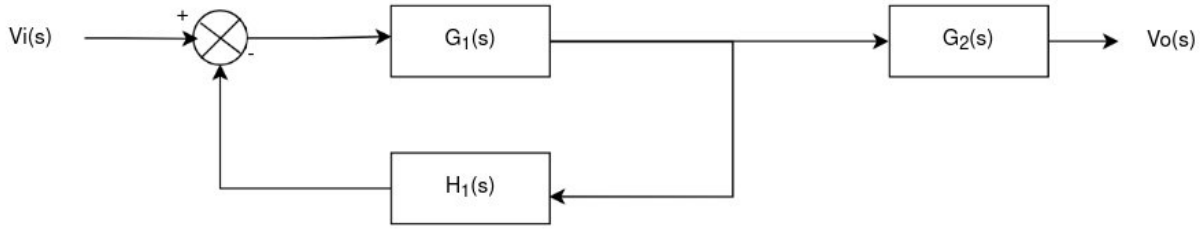


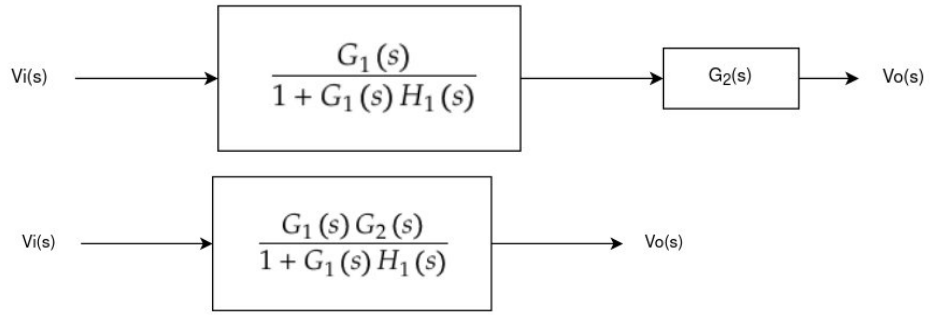


Sehingga:

$$\frac{Vo(s)}{Vi(s)} = (G_1(s) \cdot G_2(s) \cdot G_4(s)) - (G_3(s) \cdot G_4(s))$$

c.

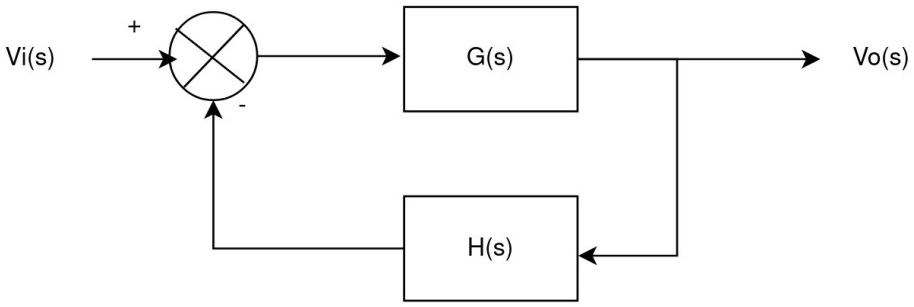




Sehingga:

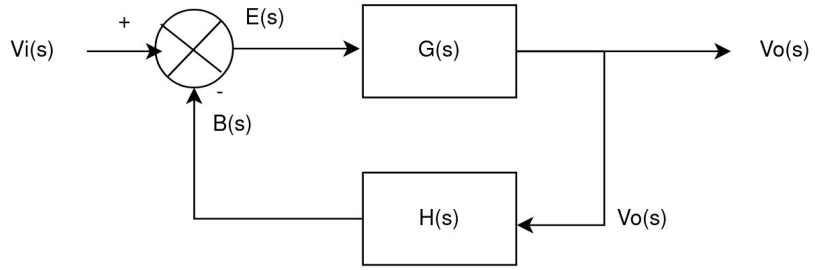
$$\frac{V_o(s)}{V_i(s)} = \frac{G_1(s)\,G_2(s)}{1 + G_1(s)H(s)}$$

3. Buktikan untuk diagram blok berikut



$$\frac{V_o(s)}{V_i(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

A. Tentukan E(s) dan B(s) pada titik penjumlah (sum):



Dari gambar, didapat bahwa:
E(s) didapat dari hasil dari titik penjumlah:

$$E(s) = V_i(s) - B(s)$$

B(s) didapat dari Vo(s) yang melewati H(s):

$$B(s) = V_o(s) \cdot H(s)$$

Subtitusikan B(s) ke dalam persamaan E(s)

$$E(s) = V_i(s) - V_o(s) \cdot H(s)$$

B. Tentukan Vo(s)

$$V_o(s) = E(s) \cdot G(s)$$

$$V_o(s) = (V_i(s) - V_o(s)H(s)) \cdot G(s)$$

$$V_o(s) = V_i(s)\,G(s) - V_o(s)H(s)G(s)$$

$$-V_i(s)\,G(s) = -V_o(s)\,H(s)\,G(s) - V_o(s)$$

$$V_i(s)\,G(s) = V_o(s)\,G(s)\,H(s) + V_o(s)$$

C. Faktoran Vo(s)

$$V_i(s)\,G(s) = (G(s)\,H(s) + 1)\,V_o(s)$$

D. Bagi kedua ruas, sehingga didapat Vo(s)/Vi(s)

$$\frac{V_o(s)}{V_i(s)} = \frac{G(s)}{G(s)H(s)+1}$$